Sound Sculpture

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ABSTRACT
This paper describes the process of prototyping, designing, testing, and assembling an interactive sound sculpture to be used in Professor Ja-Naé Duane's research in user interaction and experience design. We developed a gyroscopic MIDI interfacing technology that allows users to send rotational data to digital audio workstations (such as GarageBand or Logic Pro) to generate audio out of software instruments. Front-end designs were iterated on and tested based on this back-end technology, finding a match between the electronics and physical user interaction. These techniques have an impact in the field of audio installations and can be replicated to send any type of sensing data in MIDI form to generate sound. The result is a half-cylinder-shaped rocking device that generates sound through its oscillation on a primary axis.

1. Introduction

Music is traditionally experienced in two primary ways: actively performing or passively listening. While recorded music has made listening more accessible than ever, it has also reinforced a largely passive relationship between audiences and the musical experience. Interactive music installations present an opportunity to bridge this divide, allowing participants without formal music training to engage in the process of music creation through physical interaction. As wireless sensors and other IoT technologies continue to advance, new forms of interactive instruments can be designed to transform everyday spatial interaction into musical output.

Interactive music systems have long been explored as a way to engage non-musicians in active musical participation. Early research in interactive digital music systems emphasized the importance of designing interfaces that translate user interaction into musical expression through simple gestures or physical manipulation (Rice, 1996). One such early example is Waxman's *Digital Theremins: Interactive Musical Experiences for Amateurs Using Electric Field Sensing* (Waxman, 1990). In this installation, untrained participants use simple physical gestures to adjust the pitch of a theremin, an electronic musical instrument that is played without any physical contact. It works by using two radio frequency oscillators connected to metal antennas, where the player controls pitch by moving their right hand toward the vertical antenna and volume by moving their left hand near the loop antenna.

Within the visual arts and media art domains, sound sculptures have emerged as interdisciplinary forms that combine sound, space, and physical objects to create immersive experiences. The presence of sound fundamentally alters the perception of space and time within an installation, shaping how participants experience and move through the environment. Because sound unfolds temporally and interacts with architectural space through reflection and resonance, sound-based installations encourage audiences to explore and inhabit the work rather than simply observe it (Iturbide, 2007). Furthermore, it is also of interest to explore how sound can be activated not just through human interaction, but also through environmental forces such as wind or motion (Iturbide, 2014).

Despite the growing body of work in interactive music systems and sound sculptures, many installations remain inaccessible to audiences due to geographical distance, technological inaccessibility, or cost constraints required to build an installation. There is therefore a need for an interactive sound sculpture

that is intuitive for participants of all ages and musical backgrounds, can be set up in a variety of environments, either indoors or outdoors, and is relatively low-budget in materials compared to other existing sound sculptures.

Drawing inspiration from previous location-aware musical installations such as Masary Studios' Sound Sculpture, which used wireless blocks as spatial musical controllers, this work investigates an alternative approach using half cylinder 3D models as interactive musical objects (Masary Studios, 2016). Each half-cylindrical shape is equipped with motion-sensing gyroscopes capable of detecting angular velocity and orientation. When participants push the shape, their movement generates sound whose volume and pitch vary according to the speed, angular momentum, and displacement of the rocking motion from the circular angle.

2. Methods

2.1 Iterative Design and System Development

Our team focused on developing and refining ideas through an iterative process across both the backend and frontend for our initial prototypes. We began with a rough concept of the final experience with a clear emphasis on interaction and play, as that was what excited our group about this project. This initial direction provided a strong foundation for our decision-making and allowed us to quickly move into prototyping. Our client played an active role during this process by providing weekly feedback on early ideas in virtual critique meetings we hosted to provide updates. In particular, Professor Duane responded positively to the concept of placemaking and interactive art, which became a central focus and consistently informed our design choices as we continued iterating.

Given the original design brief, we first oriented our electronic backend to be based on the sound sculpture from Masary Studios, ideating on using ultra-wideband (UWB) modules, like the DWM1001-DEVs, with a coordinate-based system to map musical notes in physical real-time space. This approach would have allowed us to track position across an (x,y) plane and translate movement into sound based on location coordinates, however, after consulting with people like Kieran and Professor Harris who are more experienced with electronics as we did not have as much of a background with this technology, we identified several constraints: the system would be costly to implement and difficult to test within our timeline, making it impractical for rapid prototyping and iteration.

Based on this feedback, we pivoted to a more feasible approach using the MPU6050 accelerometer and gyroscope module. This shift also changed how we thought about data collection, as instead of tracking spatial position from anchors that created a global, constrained coordinate plane, we focused on capturing rotational motion data from each individual object. Each shape now is treated as its own independent unit, using its internal movement to drive interaction and generate outputs. This approach simplified the system, reducing costs and allowing us to test and iterate more quickly which better aligned with the goals of our project. We validated this direction with our client, who agreed that it paired well with the interactive goals of the project, and we proceeded with this model.

2.2 Electronic Backend and Motion Tracking Implementation

We decided to pair the MPU6050 with an ESP32 microcontroller to enable wireless data transmission. The ESP32 was chosen due to its integrated Bluetooth Low Energy (BLE) capabilities and widespread documentation. We began by prototyping on a breadboard. Using the Arduino IDE Serial Monitor, we verified that we could successfully read raw gyroscope data (angular velocity in the x, y, and z axes, measured in degrees per second). This confirmed that the sensor and microcontroller were correctly interfaced. To enable wireless data transmission, we uploaded custom firmware to the ESP32 using Arduino IDE utilizing the ESP32 BLE libraries to configure the board as a Bluetooth peripheral.

To evaluate the wireless performance of our module, we used a BLE scanner mobile application on iPhone to detect and connect to the ESP32. We measured the effective communication range by walking across the Brown Design Workshop and observed a reliable connection up to approximately 40 feet which satisfied our client's spatial requirements. Next, we developed a Python script to receive this data on a laptop using the Bleak library, a cross-platform BLE client, to scan for nearby BLE devices, identify our ESP32 module by its advertised name (ESP32_MPU6050), connect to the module, subscribe to notifications from the BLE characteristic, and parse the incoming byte data into floating-point gyroscope values. This Python script and all other firmware and client-side code were developed using the assistance of the AI chatbot ChatGPT due to our time constraints and dearth in background embedded systems coming into the project (OpenAI, 2026).

Although this successfully streamed real-time data, we observed that even when the gyroscope was stationary, the readings exhibited small fluctuations (typically ±4 degrees per second). To address this, we implemented signal processing techniques directly in Python, as recommended by Professor Harris, including deadband filtering and exponential smoothing to eliminate minor noise. With this clean motion data available, we developed a second script to translate gyroscope input into sound. This script mapped motion features to MIDI parameters as follows: overall motion magnitude (computed as $\sqrt{(gx^2 + gy^2 + gz^2)}$) controlled volume and directional components (gx, gy, gz), and controlled pitch modulation. To route MIDI data into audio output, we used macOS's Audio MIDI Setup to create virtual MIDI buses (IAC Driver), which were routed into GarageBand, where they controlled software instrument tracks in real time.

Finally, we explored scaling the system to multiple prototypes. Using three ESP32-MPU6050 units, we modified our Python code to connect to multiple BLE peripherals simultaneously. Each device maintained its own calibration, filtering, and MIDI mapping pipeline. One challenge we encountered was that GarageBand does not support the clean separation of multiple MIDI inputs across tracks, causing all gyroscopes to influence all instrument tracks. To address this, we implemented multiple virtual MIDI buses (IAC Bus 1, Bus 2, Bus 3), routing each device to a separate bus. However, due to GarageBand's limitations, our step was to transition to Logic Pro, which allows explicit mapping of MIDI inputs to individual tracks, enabling independent control of multiple instruments.

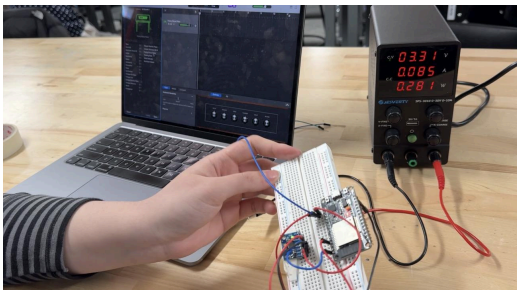


Figure 1. Initial working electronics module with GarageBand

2.3 Physical Prototyping and Form Exploration

To explore how form could support interaction, we developed a series of physical prototypes designed around movement and rotation. In total, we created five distinct forms: a spinning-top mockup, a cube based on the original design brief, a half-cylindrical sphere, a hexagonal sphere, and a half-cylinder. Each prototype was intentionally designed to encourage different types of motion, allowing us to test how physical manipulation could drive interaction. With our electronics backend in mind, the prototypes were built to take advantage of the gyroscope-based system, where rotation and movement serve as primary

inputs. This allowed us to evaluate how shape, balance, and interaction influence the overall user experience.

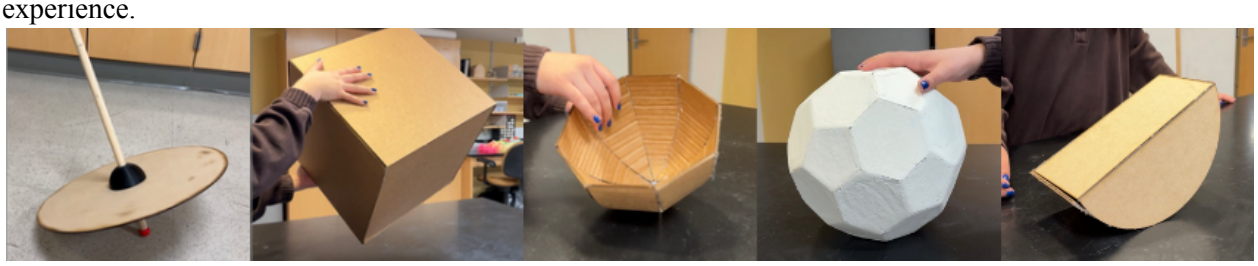


Figure 2. Five prototypes designed around movement and rotation

Our physical iteration process depended on meetings with Professor Duane, where she outlined the critical design criteria for this project, and also considered the backend gyroscope technology that would depend on specific motions in the front-end. Our final criteria included movability, the ability to be used indoors and outdoors, a rotational element that continues to rotate/move after use, and a built process that is replicable and repeatable. We then explored different solutions to these criteria through explorations of interaction. We had to consider many physical/mathematical constraints and relationships to get these prototypes to function, especially for the spinning top, where the weight/dimension ratios had to be precisely calculated to ensure continued rotation after use, and also for the geometric considerations of other shapes we explored.

To prototype these forms physically, we used a range of materials easily found in the BDW, such as cardboard, hardwood, wooden dowels, PLA (3D printing) and resin (resin printer). Early prototypes, specifically the spinning top, were made using pink foam due to its availability, but its difficulty to work with quickly showed us the limitations of this material, especially when the spinning top required a high level of precision for stable rotation.

The spinning top in particular posed significant challenges for use due to its sensitivity for dimensional accuracy and weight distribution. Our initial attempts involved tracing large circles on cardboard with a compass and cutting them out by hand, but this process was quite inefficient and inconsistent. After researching the geometry behind a spinning top, we found that the diameter of the base and the height of the shaft must be equal, and even slight irregularities in the shape of the circle introduced imbalance that caused the top to wobble and fall. To address this, we switched to laser cutting the circle, which resulted in a slightly smaller diameter, but it was precise and easily reproducible. We laser cut hardwood circles for the base, used wooden dowels for the shaft, and designed a 3D printed cap for the dowel to allow for smooth rotation, resulting in a significantly more stable and functional prototype.

We also faced challenges in constructing the spherical forms. We initially considered paper-mache over inflatable beach balls, but we figured that structure would lack the rigidity needed for repeated, constant interaction; one of our criteria was to make sure the final shape is strong enough to withstand people using it without fear of it breaking. Instead, we switched to scoring and folding cardboard to create curved surfaces, allowing us to create lightweight and sturdy spherical objects. We explored several spherical/cylindrical shapes with this approach, including a geodesic sphere made of 5 separate panels, a hemispherical shape made up of 8 gores, and hybrid forms such as half-cylindrical forms that would rock back and forth.

Across all of our prototypes, we prioritized designs that could withstand rotation, oscillation, or rocking motion after user interaction, aligning well with our gyroscope-based electronic system. This required very careful attention to balance, weight distribution, and points of friction to ensure smooth and continuous movement. At the same time, each form needed to accommodate our internal electronics, so it

was beneficial that most of our prototypes were modular; they were composed of multiple panels or subshapes that could be easily assembled and disassembled, allowing the electronics to be easily placed inside of the main shape.

3. RESULTS AND DISCUSSION

3.1 Final User Testing and Iteration

Similar to our methodologies for building and iterating on our prototypes, we had to test both our front-end physical designs and interactions, our back-end electronics and functionality, and then the relationship between them in the experience of using the sculptures to generate sound.

For our front-end testing, we iterated up from our low-fidelity prototypes through our initial design reviews with Professors Harris and Manfredi, as well as our design TA, Maia, which led us to explore further rotational objects after showing them our research on a pendulum hanging from the ceiling that generates sound. One of these tests included hanging a pink-foam machined ball from the ceiling of our classroom and swinging it around, allowing us to notice the physical constraints of the string supporting the ball, as well as the need for an apparatus that the balls would hang from if we proceeded with that idea. After deciding on gyroscope technology, these tests led us to iterate towards medium fidelity prototypes that emphasized rotation and movement in a more intuitive and less involved way.

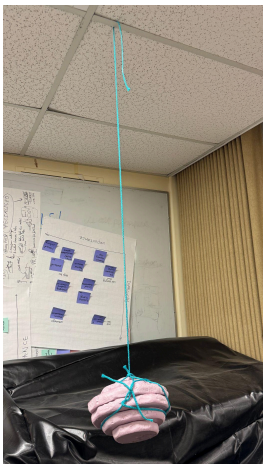


Figure 3. Swinging pink-foam ball prototype

In terms of our back-end sound, Initially, pitch was mapped across a full chromatic range, but this produced dissonant results. To improve musical coherence, we constrained pitch selection to a G Major pentatonic scale, ensuring that all generated notes were harmonically compatible regardless of input. This significantly improved the aesthetic quality of the sound.

Initially, the system was powered by an external 3.3V supply on a breadboard. However, this design was very bulky and difficult to transport reliably for a user. To make the device self-contained, we transitioned to battery power. Since commonly available lithium batteries output ~3.7V, we incorporated a voltage regulator module into our design to step the voltage down to 3.3V for safe operation of the ESP32. We also reduced the footprint by switching from a full-size breadboard to a mini breadboard, and eventually soldered the components onto a perfboard for a more robust and compact assembly. This compact system allowed the prototype to be physically much more seamlessly, marking a major milestone in the project.

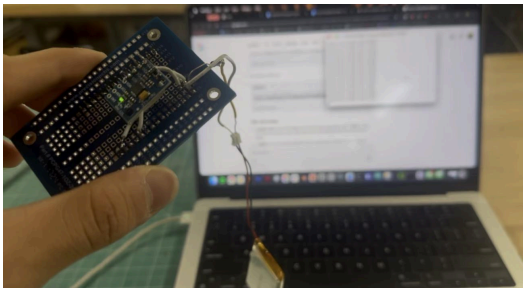


Figure 4. Final prototype electronics module

3.2 Final Prototype

Our final physical prototype consists of two ⅜-inch plywood semicircular side panels connected by nine wooden dowels spanning the curved perimeter, with an additional rectangular wooden member spanning across the top structure for support and stability. The exterior faces of the prototype were painted a deep brown, while the dowels were painted white and the interior surfaces and edges were painted a bright cerulean blue. Two white 3D printed capsules are sealed with silicone glue to the underside of the rectangular panel: one to house the electronics and the other for symmetry. The two capsules have curved edges that smooth into the surface of the wood to create a seamless visual.

The exterior brown finish was selected in consultation with our design teaching assistant, Maia, in order to allow the sculpture to visually blend into outdoor environments such as the Main Green at Brown University without disrupting the surrounding landscape. In contrast, the bright blue interior was intentionally chosen to create a visual invitation for interaction. Feedback from peers within the Brown Design Workshop suggested that the vibrant interior surfaces contributed to the sculpture feeling more playful and approachable, encouraging users to physically engage with the object in a manner similar to interacting with a toy.

The final prototype successfully integrated the electronic and physical systems into a cohesive interactive artifact. Users were able to rotate, tilt, and manipulate the sculpture while the embedded gyroscope system generated corresponding real-time musical output. The resulting interaction demonstrated the viability of using motion-driven embedded electronics to create an intuitive and expressive sound-generating experience.



Figure 5. Final prototype physical form

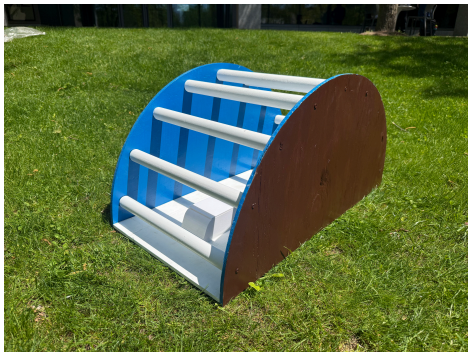


Figure 6. Final prototype flipped upside down

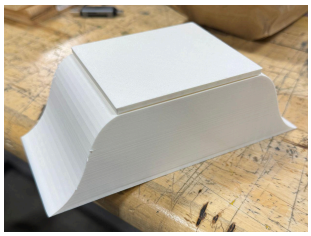


Figure 7. Final prototype electronics capsule

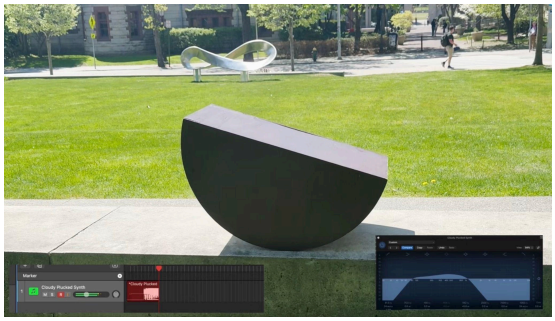


Figure 8. Final prototype with display of backend gyroscopic data and Logic Pro visualization

4. LIMITATIONS AND FUTURE DIRECTION

4.1 Limitations

One major limitation of this project was the restricted development timeline which constrained the extent to which we could explore multiple interaction experiences and create additional prototypes. Though we initially planned to have three working modules by the end of the course, we ran into soldering issues and a lack of suitable prototyping board for our use case and were only able to have one main functional prototype. Our broader intention was to create multiple sculptures capable of collaboratively generating layered musical compositions as a "symphony" which would have more fully aligned with Iturbide's discussion of sound installations as immersive environments (Iturbide, 2007).

Initially, we intended to fabricate a continuous curved wooden enclosure spanning the underside of the sculpture in order to create a more seamless final form. However, due to material and tooling constraints, we instead utilized exposed dowel structures to achieve the desired geometry. Finally, the current electronic implementation still retains several usability limitations associated with prototyping, including the need to manually press the ESP32 reset button to restart the system and the use of non-rechargeable batteries.

4.2 Future Directions

As noted above, we would implore future researchers to scale our pilot sculpture to multiple simultaneously functioning sound sculptures to create a "symphonic" experience for users. Furthermore, while our original concept involved real-time spatial location tracking using multiple sensors distributed throughout a room, budgetary and time constraints ultimately led us to pivot to the gyroscope-based system. While our team ultimately found the solution successful, we believe a further exploration of a system combining both the gyroscope-based system and real-time location tracking system like that of Masary Studios is worth exploring (Masary Studios, 2016)

Future work could also explore additional gesture-based interaction methods inspired by Waxman's *Digital Theremins*, which demonstrated how simple physical gestures can allow non-musicians to intuitively participate in musical performance (Waxman, 1990). From a fabrication standpoint, future versions would incorporate smoother curved structures, rechargeable battery systems, and externally accessible power switches to improve long-term installation reliability.

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